

Evidence-Based Hot-Path, Detection & Capacity Plan

What the firewall actually costs, what it actually detects, and what \$5,000 / month actually buys — measured, not derived.

≈5.9 ms

P50 ADDED LATENCY, ≤1,024 TOK, IN-PROCESS L4 GPU — 74% DERIVED, NOT MEASURED

590–1,064

RPS ON \$5K/MO — BOUND BY GPU QUOTA, NOT BUDGET

1.99%

PER-REQUEST FALSE-POSITIVE RATE, NEVER PREVIOUSLY STATED

Verdict changed 2026-08-29. All architectural constraints were removed (GPU/CPU/RAM open, budget unchanged). That lifted the GPU prohibition — which was a signed lock, not an evidence-based conclusion. **≤12 ms is reachable** with the classifier in-process on an L4 node. The CPU-only analysis below remains correct and is why.

On CPU only: NOT ACHIEVABLE — over-determined three ways

Constraint	Measured	Budget	Over by
Semantic classifier — PG2-22M INT8 ONNX, seq 512	159.2 ms	5 ms	32×
Tier-1, verdict-identical rewrite, 1024-token band	63.4 ms	0.50 ms	127×
Cost of 100k RPS, coalesced SSE, best case	\$309,620/mo	\$5,000/mo	62×

And the finding that neither hardware nor budget fixes: the deterministic layer was never measured. It detects **0 of 10** paraphrased injections while hard-blocking **5 of 5** benign prompts containing inline code.

EVIDENCE BASE

18 agents across two workflows — 7 evidence agents, 1 synthesis, 5 adversarial reviewers (CLAUDE.md Phase 3), 1 devil's advocate (Phase 4), 3 capacity/removal agents, 1 adjudicator. ~4.1 M subagent tokens, 1,178 tool calls.

Provenance tags used throughout. **M** measured this session · **R** repo bench corpus · **V** vendor-published, URL + date · **I** independent third party · **D** derived arithmetic, assumptions stated · **NF** searched, not found.
2026-08-27 · branch ansh · PLAN ONLY — no code, config, or infrastructure change is authorised by this document.

1 · The finding that matters

The plan was about to recommend shipping Tier-1-only as the default posture and certifying 12 ms for it. Nobody had ever measured what Tier-1-only detects.

Measured against the real `InputScanner._scan_prompt_sync` at HEAD:

Probe	Result	Meaning
Bare English prompt injections, same family as the shipped regexes	3 / 10	detected
The same injections paraphrased with disjoint trigger vocabulary	0 / 10	detected
Benign prompts containing inline-backtick code	5 / 5	hard-blocked, confidence 1.00
Ordinary agentic tool descriptions	3 / 12	hard-blocked, confidence 1.00

Root cause. `ATTACK_PATTERNS["command_injection"]` contains the literal pattern `r"``[^\`]+\`"` — **any backtick pair**. `command_injection` and `data_leakage` are excluded from the explanatory carve-out ( `scanner.py:104` , tested at `:1201` ), so the block is terminal at `:1212-1219` , `tier_1` , and never reaches Tier-2.

The recommended default posture is near-blind on the primary threat and hostile to ordinary developer traffic, simultaneously.

This inverts the project

It is not "12 ms vs 119 ms." It is: *the product's detection value lives almost entirely in the layer that costs 100× the SLO, and the layer that fits the SLO has never been measured and does not work.*

Consequence for the Tier-1 rewrite

A verdict-identical rewrite is *achievable* — 0 verdict diffs across 44 `_scan_prompt_sync` inputs, 1,432 `_segment_token` tokens, 28 `_fuzzy_scan` inputs `M`. It is a 32× win. **But byte-identical verdicts is now the wrong goal:** it is a gate that faithfully preserves a detector measured at 0/10 paraphrase recall and 5/5 false-positive on inline code.

Tier-1 option	Cost @1024 tok	Verdicts
Verdict-identical rewrite	63.4 ms <code>M</code>	preserved — <i>including the defects</i>
Change the detector	floor 0.34–0.54 ms <code>M</code>	different — must be scored against a corpus that does not yet exist

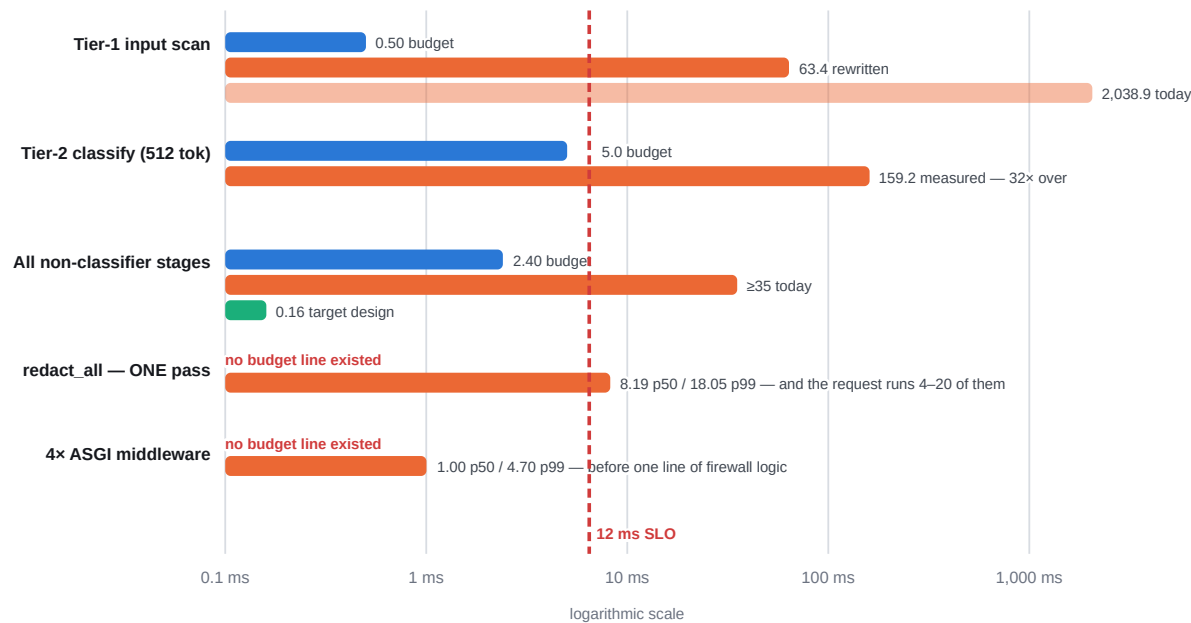
You cannot choose between these without a detection corpus. That is **Gate 0**.

The instrument is also broken. `ttft_ms` and `duration_ms` both derive from `provider_start_ts` (`pipeline_trace.py:120-131`); `total_latency_ms = duration ( :738, :778 )`; the bench computes `addon = total - model_output` . **Therefore `addon` = TTFT.** On streaming requests the benchmark measures the customer's model, not the firewall — so the 12 ms gate passes unconditionally and Phase 1's "stop and revise" gate cannot fire.

2 · Measured latency vs budget

Every bar is measured in-process against the real modules. The budget column is what the superseded plans assumed.

Budgeted D Measured M Target design M



Measured on Intel Xeon Platinum 8581C, Python 3.12.13 (matching the gateway Dockerfile), 1024-token band. **Nothing here was measured on the target ARM SKU** — every millisecond is x86.

Why the classifier cannot be rescued

Cause	Evidence
The model is 12 layers , not 6	R2's spec was wrong; every downstream estimate inherited it M
Batch-4 is net negative at the optimal thread count (0.92× at 8 threads)	The apparent 1.49× win exists only at 16 threads — compensation for wasting cores on one small graph M
INT8 does not help — 1.27× over FP32, and 1.73× <i>slower</i> at batch-4	ONNX Runtime's fusion optimiser's <code>MODEL_TYPES</code> dict has no deberta entry → no fused Attention → no QAttention . Architectural, not a tuning knob V
Two costs never budgeted	SentencePiece tokenisation 4.3 ms/512 tok; and a 16-core host cannot supply the FLOPs for even 85 RPS at 1024 tokens M

Useful for the real design: 8 threads (= physical cores) is optimal; **16 threads is 25% slower with 2.4× worse p99**. SMT actively harms this workload — the cliff must be re-found on Axion.

3 · The embedding + trained-head alternative — and why it fails

A static embedding with a trained classifier head looked like the escape route: ~500× cheaper than a transformer, with published precedent (NemoGuard-JailbreakDetect, F1 0.9601, is a classifier head over embeddings — not a transformer judge). The devil's advocate trained the head that nobody else had trained.

Window	≈ tokens	AUC same-family	AUC disjoint-vocab	Recall @1% FPR
16 words	~21	0.9995	0.9498	0.377
48 words	~62	0.9566	0.7708	0.060
200 words (deployed)	~260	0.7397	0.6017	0.028

AUC 0.60, recall 2.8% at the deployed window. The mechanism is **dilution**, not order-invariance — it works at 21 tokens and collapses at 260. And the arithmetic that kills it outright: **51 windows × 1% per-window FPR = 40% per-request FPR.**

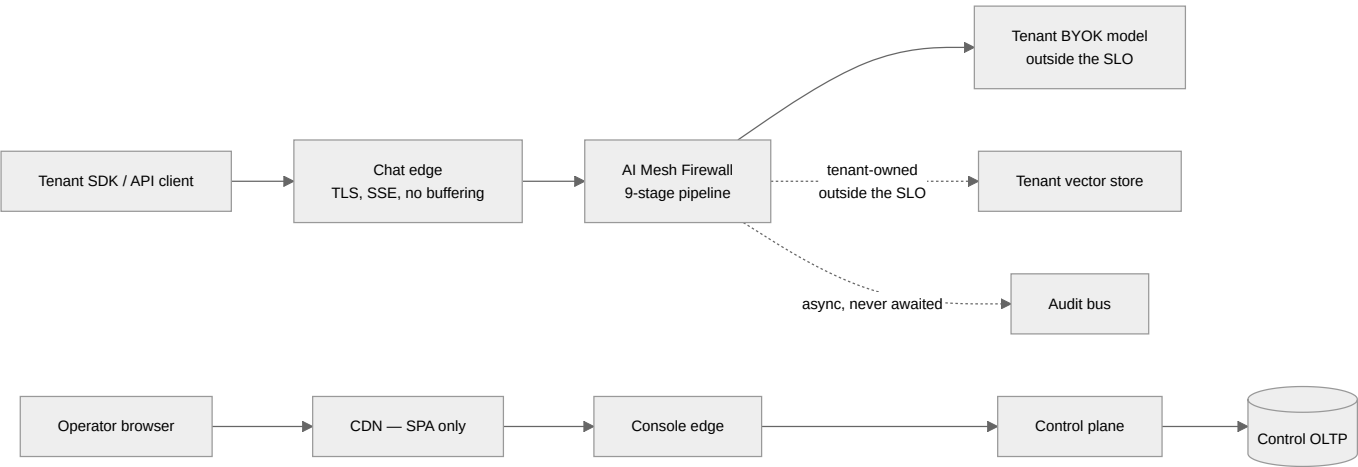
Cost was never this design's problem (~1.5–2 ms at the band). **The multiple-comparisons FPR budget is.** A cost-only gate admitted it; a quality gate excludes it. This is why *L2 must name a model class with a GFLOP-per-window entry gate and a published recall floor, evaluated at the deployed window size with a per-request FPR budget.*

The lever that was never evaluated

GPU was rejected using an AWS price in a GCP-locked plan. NF — no GCP accelerator rate appears anywhere in the bill of materials. GCP `asia-south1` has `nvidia-l4` in zones `a/b/c` with on-demand quota 8, and `g2-standard-4` machine types. This is the *only* lever that closes the classifier gap outright, and it has never been priced on the correct list.

4 · System context

Two products, two edges. The chat path and the operator console must never share an edge.



The contract boundary is **who owns the endpoint**. Tenant-owned round trips — the BYOK model and the tenant's vector store — sit outside the latency contract on the same principle.

The per-path contract

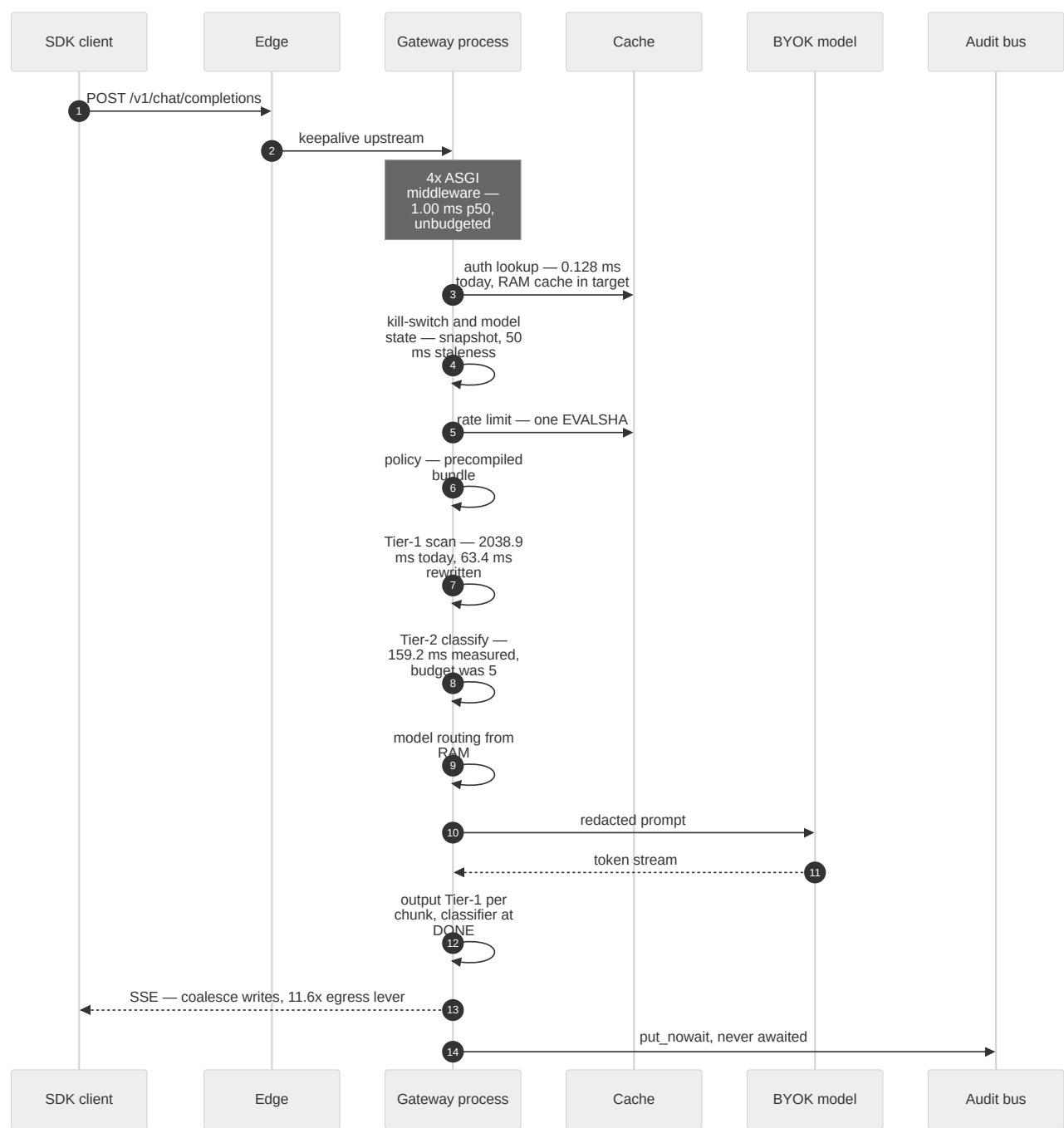
All four paths meet a 12 ms tax in a **Tier-1-only posture** — the posture Gate 0 must first prove is worth shipping.

Path	Firewall tax, target	
	posture	Note
Chat	2.39 ms	a sum of <i>unbuilt targets</i> mis-tagged as measured; honest range 3.0–5.5 ms p50 , p99 unestablished
MCP	1.5–3.5 ms	inherits the chat scanner
RAG	0.5–0.6 ms	wall clock is ≥124 ms and never lower — Pinecone has no in-region endpoint
Vector	0.7 ms	—

The contract, stated exactly: " $\leq X$ ms of firewall tax + N declared tenant round trips, $N \leq 2$." Today $N = 3$ on Pinecone — **and the third one is ours**.

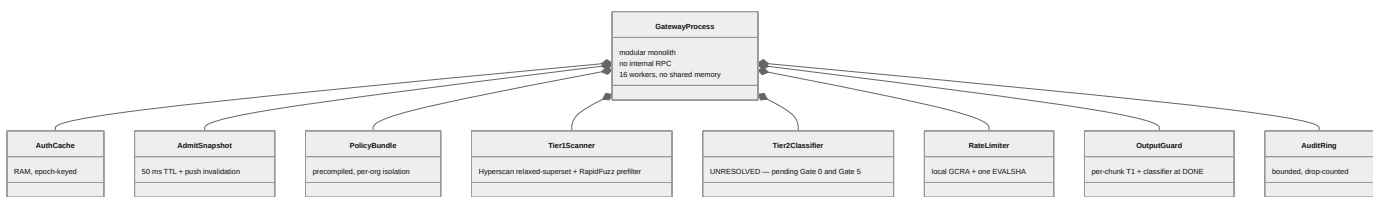
The useful collapse: requiring the SLO on all four paths does not multiply the problem by four — **it collapses it to one**, because all four share the scanner. That one question has a measured answer, and it is currently *no*.

5 · The nine-stage pipeline, with measured cost



Stage costs are measured in-process **M**. The two stages with no budget line at all — ASGI middleware and tokenisation — are among the largest.

Gateway process — component view



Per-worker state lies at scale. `compute_sizing(16.0, 64 GiB) → workers=16` with no shared memory **M** — so a breaker configured at `MIN_CALLS=5` is effectively 80. Every cache line in the tracker carries a "worker-count multiplier applied? Y/N" column.

6 · Cost — egress dominates, compute does not

Egress plus load-balancer data processing is **94.6%** of the best realistic case at 100k RPS. Compute is **5.1%**.



Assumptions: `c4a-highcpu-16` 1-yr CUD, ≤60% CPU, ×1.5 AZ, min 3 hosts HA, no Cloud NAT, Premium Tier egress to Asia, 70% streaming, 400-token answers, 250 RPS/vCPU, no Cloud Armor, no Cloud Logging at request rate. The gzip series is D, not measured.

The reframe. Making the code faster buys nothing. **Making it emit fewer bytes buys 11.6×.** Both are code changes — but they are different code, and only one of them was in the old plan.

The highest-return single fix

The terminal SSE frame carries the full `pipeline_trace`: **41,424 bytes, with 21 copies of the prompt and 7 of the response**, on every *allowed* request, streamed and non-streamed, with no config flag. `_scrub_trace_for_client` (`main.py:723`) is called from **exactly one place** — `:955`, the blocked path. This is **24% of the entire \$5k budget today**.

The number nobody agrees on

Source	RPS/vCPU	Per 16-vCPU host	Tag
MASTER D4's 12-host cell for 100k	520	8,320	no support anywhere — strike it
MASTER D4's stated header	250	4,000	D
Synthesis planning band	9–25	150–400	D
FLOP wall, PG2-22M @1024 tok	0.68	10.8	D from M
Today, measured, @1024 tok	0.3	~5	R

A 1,700× spread on the second-largest bill line. **$N \approx 85$ RPS/host is a 32–64-token number**; at the 1024-token band today's host does ~5 RPS.

7 · Competitive positioning

Every published gateway benchmark is inflated by methodology. Knowing exactly how is what makes an honest claim possible.

Gateway	Published	The caveat that matters
TrueFoundry	+3–5 ms (<i>blog</i>)	+7 ms in the docs, +12 ms with tracing ; 350 RPS/vCPU; mock upstream; auth + routing only, zero security scanning
Kong	96k RPS, p99 9.75 ms	16 vCPU; <i>"no policies, like caching or API key based authentication, were configured"</i>
LiteLLM Rust	0.05 ms	<i>"no logging callbacks, spend tracking, or persistence enabled"</i>
Bifrost	11 μs	<i>"excludes upstream response time"</i>
Envoy AI Gateway	NF	no published data-plane benchmark exists

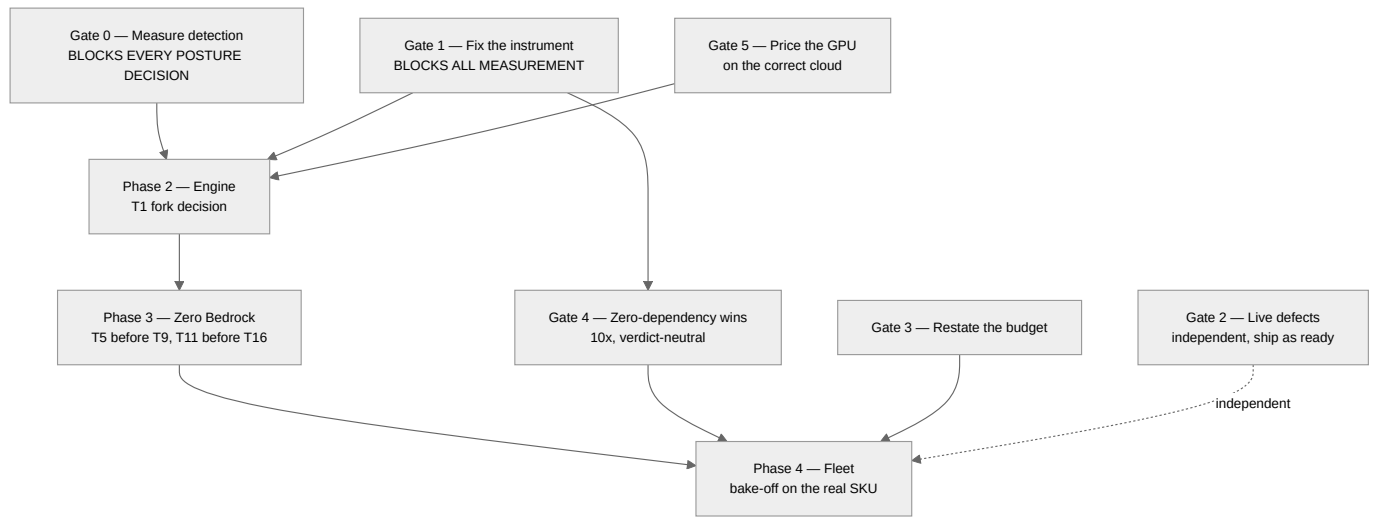
The finding that governs every comparison: *"None of the published overhead numbers are measured with streaming on."* Kong's own documentation states streaming cannot combine with response-phase plugins at all.

The defensible claim is therefore not RPS parity. It is latency parity while running a security pipeline none of them run, measured with streaming on — which requires fixing the instrument first.

Capacity, as three separate claims

Claim	Statement	Status
C-A design ceiling	No fixed bottleneck below 100k RPS; every tier scales horizontally with a named binding constraint	achievable as a design property
C-B provisioned N	What \$5,000/mo actually buys, measured, with HA	≈ 1,000 RPS with coalescing (\$4,866/mo). ~122 RPS as the code stands. In-flight is never binding at this budget.
C-C competitive parity	Added latency comparable to industry AI gateways	blocked on the instrument

8 · Execution — gates and dependencies

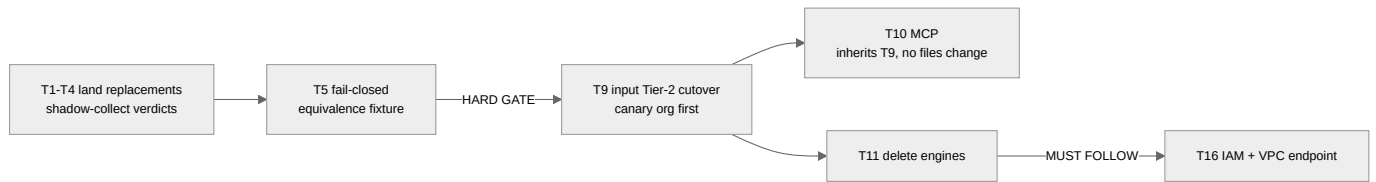


The single most important sequencing rule: Gate 1 before any measurement, Gate 0 before any posture decision. Reversing either produces a confident, published, wrong answer — the failure mode this corpus caught twice already.

Live defects — not performance work

ID	Defect	Why it is severe
S-1	PolicySync partial cold start	A subset of tenants is unenforced forever , /health green, no reconciliation. Workers × nodes = independent scans, each able to stop at a different point.
S-2	Cross-tenant config inheritance	build_redis_key() returns firewall:config:default when org-less, so an un-synced tenant inherits the platform-default posture . The fix exists and was applied to MCP but not to chat.
S-3	--worker-connections silently inert	No admission control anywhere. Raising timeoutSec first converts a routine upstream stall into a metastable failure.
S-4	scan_degraded dead on the output path	On breaker-OPEN, streams ship Tier-1-only, fail-open, while telemetry attests they were scanned . A false attestation in a compliance record.
S-5	Streaming "block" is a truncation	260 of 800 characters delivered, then an error frame. An operator who selected block did not get one.
S-6	Telemetry drops evidence	Whole batch discarded on Redis error — precisely during the failures the plan expects .

Zero-Bedrock ordering — two laws



T5 before T9 — never remove the only fail-closed control without proving its replacement. **T11 before T16** — revoking IAM while code still calls Bedrock converts a fail-open path into a per-request `AccessDeniedException` storm.

10 · Best achievable — all constraints removed

5 proof agents → assembler → independent verifier. The audit returned **15 confirmed, 4 overstated, 3 invalid**. This is the post-audit position.

Prompt band — stated in <i>tokens</i> , as it must be	windows	p50 added	tag
≤ 512 tok	1	≈ 4.1 ms	D
≤ 1,024 tok	2	≈ 5.9 ms	D
10,000-char deployed cap, prose	4	≈ 10.8 ms	D
10,000-char deployed cap, markdown / code	7	≈ 17.2 ms	D

p99 **18.3 ms** at 1,064 RPS; **13.3 ms** at 400 RPS D queue-simulated. Convergence band on the 2-window p50: **5.4–6.9 ms**. **74% of the p50 has never been measured on any GPU we control.**

The architectural unlock

G2 bundles the GPU and the vCPUs in one VM M — confirmed three ways. So the classifier runs **in-process with zero network hop**. Every alternative topology starts at **≥16.1 ms RTT**, which is exactly the hop tax MASTER D1 said a 12 ms budget cannot absorb. **The GPU decision does not pay that tax — it dissolves it.**

4 × **g2-standard-24** in asia-south1, zones a/b/c, 3-year CUD — 96 vCPU + **8 × NVIDIA L4**.

The runtime is not a tuning knob. ORT 1.29.0's MODEL_TYPES has no deberta , **and fusion is a graph-level pass applied before execution-provider assignment** M — so the ORT CUDA EP would run *the same unfused graph* on the GPU. Native TensorRT is mandatory, served by Triton with both GPUs behind **one shared queue** (worth **8 ms of p99** versus 8 single-GPU nodes, at identical price).

Tier-1 rebuilt — all measured

Change	From	To
One Hyperscan relaxed-superset gate over all 174 patterns	21.97 ms	0.160 ms
Native segmentation	14.72 ms	0.434 ms
Dictionary admission filter (DICT \ D_seg)	—	16.6× less DP work, 0.00% recall loss by construction
Injection patterns demoted from block to signal	—	fixes the measured 5/5, 7/7, 3/12 false positives

The RPS ladder — binding constraint named at each rung

Posture	Fleet RPS	Binding constraint
A — 100% semantic coverage	1,064	NVIDIA_L4_GPUS = 8 quota M — not the budget
A' — quota raised to 14 L4	1,862	budget, at \$1.82/RPS-mo of GPU
B — risk-gated coverage ≥26%	4,151	egress + LB
C — deterministic only	7,200–9,400	egress + LB, <i>contingent on an un-load-tested rewrite</i>
D — BYOK provider out of region	1,731	egress — prompt leg becomes 65%

The cost correction that moved the answer. Linearising the old \$4.866/RPS-month put this design at 346–511 RPS. With the trace frame deleted, SSE coalesced, gzip applied and the provider in-region, the *marginal* cost is **\$0.449/RPS-month** — a **10.8× correction**. Egress stops binding until ~4,150 RPS, so the GPU quota binds first. CPU never binds: 26,592 RPS of CPU capacity against 1,064 of GPU.

11 · What the audit invalidated

The verifier's job was to audit the proof ledger, not redo the analysis. It found no GPU number mis-tagged as measured — but three claims did not survive.

11 · "\$1,262 unspent, nothing left to buy" — INVALID

A hot-path BOM presented as a total BOM. **Cloud Logging is unpriced** — and its price page was sitting in the assembler's own evidence directory.

Missing line	Evidence it is real	Cost at 1,064 RPS
Cloud Logging	\$0.50/GiB V	200 B/req → \$232 · 1 KiB → \$1,290
MongoDB (live telemetry sink)	13 refs + motor driver, in compose	NF
ChromaDB, persistent disks, Artifact Registry, Monitoring	in compose / required	NF

At 2.758 billion requests/month, a security firewall must retain a per-request verdict record. At a realistic 1 KiB audit line, Cloud Logging alone is **\$1,290/mo** — more than the entire declared headroom — putting the plan at **~\$5,028 and over budget at its own recommended operating point**. Avoidable via Log Router exclusions or a GCS sink at \$0.02/GiB, but that is a decision to be *made and stated*, not omitted.

12 · No per-request FPR — the biggest finding

The design quotes PG2-22M's **88.7% recall @ 1% FPR** V. That operating point is **per 512-token window**, and Meta's own model card prescribes an OR over segments. So per-request FPR = $1 - 0.99^W$:

windows	per-request FPR	false ENFORCE-blocks/s @1,064 RPS
1	1.00%	10.6
2 — the headline band	1.99%	21.2
7 — deployed cap, markdown	6.79%	72.3

This is structurally the same error that correctly killed the potion-8M design (51 windows × 1% = 40% per-request FPR). Applied to its own recommendation it survives only because W is small — and it was not stated at all. In an ENFORCE posture, 2% is a product-defining number.

Two further omissions: the model card publishes **multilingual AUC 0.942** against 0.995 English — the operative figure for a Mumbai deployment — with **no multilingual recall@1%FPR published** NF; and the 88.7% is on Meta's *private* benchmark, against an 8–16 point AUC collapse under leave-one-dataset-out in our own citations.

13 · The band must be stated in tokens

Measured chars/token is **3.04 (markdown) / 6.28 (prose)** M — never the 4.00 rule of thumb. 1,024 tokens is **3,158 characters** of this corpus, not 4,096. And the **deployed cap is 10,000 characters** (scanner.py:74) — so the 6 ms figure describes a *sub-band* of the deployed configuration, presented as the configuration.

The one sentence that can be said to a customer

"A full deterministic scan plus a 22M-parameter transformer injection verdict on **every** request — synchronously and block-capable before any byte reaches the provider — at a projected **4–6 ms p50 / ~18 ms p99** of added latency for

*prompts up to 1,024 tokens, and **590–1,064 RPS** on a single-region \$5,000/month footprint; every GPU figure is derived from NVIDIA's published TensorRT measurements of this exact architecture and **none is yet measured on our own hardware**. We cannot yet quote a false-positive rate: the classifier's published 1% FPR is per 512-token window, which compounds to ~2% per request at two windows and ~7% at our deployed 10,000-character cap."*

Everything before the semicolon is strong. Everything after it is why this is not yet a product claim. **A defensible 10 ms with a published per-request FPR beats an indefensible 6 ms with none.**

The three actions, reordered by what the audit changed

1. **Measure the production prompt-length distribution in tokens.** One day of telemetry. It sets the window count — which sets the latency headline, the RPS headline and the per-request FPR *simultaneously*. Cheapest high-value action in the workstream.
2. **Run the `trtexec` bake-off** — \$7, 30 minutes — sweeping `--optShapes` at $W = 1, 2, 3, 4, 7$. It must settle: is window-batch-1 inside 2.7–4.0 ms; where throughput saturates (the 490 vs 1,580 windows/s spread is 3.2× and directly sets \$/RPS); **p95/p99 over 1,000+ iterations** — **no tail data exists anywhere**, and a 72 W card under sustained load is exactly where tails appear; and whether DeBERTa INT8 works on TensorRT at all `NF`, currently a coin-flip worth 2× throughput.
3. **Publish a per-request FPR budget before the demotion ships**, scored at the deployed window count and including multilingual.

12 · Complete cost matrix

Posture A — 4 × g2-standard-24 , semantic scan on every request, 1,064 RPS, asia-south1, 3-year resource CUD. Volume basis: **2,757,888,000 requests/month**. This corrects the assembler's \$3,737.58, which the audit found was a hot-path BOM presented as a total BOM.

Service	Spec	\$/mo	Why	Tag
COMPUTE				
GCE g2-standard-24 × 4	96 vCPU · 384 GB · 8 × L4 · zones a/b/c	2,736.76	The whole architecture. G2 bundles GPU <i>and</i> vCPU in one VM → zero network hop vs ≥16.1 ms RTT for any split. 8 L4 = the entire regional quota.	V
DATA				
Cloud SQL PostgreSQL	Enterprise HA, 4 vCPU / 16 GiB / 100 GiB	273.91	Orgs, keys, policies, users. Control-plane only — chat path opens zero DB connections. HA because a policy-store outage is a fail-closed outage.	V
Memorystore Valkey	HA, 2 × 13 GB highmem-medium	175.20	The one shared-state call per request. HA because the kill-switch fails closed — a single-node failover is a full-product 503.	V
Persistent disks	650 GiB pd-balanced	65.00	Boot volumes + the two self-hosted stores. Balanced not SSD — neither is on the hot path.	D
PLATFORM & NETWORK				
GKE	1 regional cluster fee	73.00	Per-fleet autoscaling, multi-zone, GPU node pools. Flat fee.	V
Artifact Registry	~20 GB images	2.00	Gateway + TensorRT runtime images.	D
Regional external ALB	forwarding rules	18.25	TLS, SSE pass-through, ≥350 s timeout. Regional — the global ALB is Premium-only, and Standard Tier saves 18% on egress.	V
Egress + LB data processing	4,197 GiB, Standard Tier, banded	460.46	Response bytes to clients + prompt bytes to provider. Priced through the real band table, not linearised .	D
OBSERVABILITY				
Cloud Monitoring	Managed Prometheus, cardinality-capped	150.00	Must use GMP — standard ingestion is \$0.2580/MiB, which at naive per-request cardinality exceeds the entire budget. Risk line .	D
Audit log → GCS	~400 GiB compressed	20.00	A firewall must retain a per-request verdict record. GCS, not Cloud Logging — see below.	D
CO-LOCATED — \$0 INCREMENTAL COMPUTE				
MongoDB · ChromaDB · control plane · workers · MCP broker · frontend · nginx · PgBouncer	on spare vCPU	0.00	CPU never binds : 26,592 RPS of CPU capacity against 1,064 RPS of GPU — ~25× spare. All of these fit in the headroom the GPU quota strands.	D
DELIBERATELY NOT BOUGHT				
Cloud NAT	—	0.00	\$0.045/GiB both directions would make the otherwise-free 87 KiB provider response billable.	V
Cloud Armor	—	0.00	\$15,180/mo at these rates — 3× the entire budget .	V
ClickHouse · Kafka · BigQuery · Vertex Vector Search	—	0.00	Not needed at this volume; each is a new operational surface.	—

TOTAL	\$3,974.58	Headroom to \$5,000: \$1,025.42 · \$1.44 per million requests
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The one decision that breaks the budget

Audit destination	Volume	\$/mo	Verdict
GCS, compressed — recommended	~400 GiB	20	fits, \$1,025 headroom
Cloud Logging @ 200 B/request	526 GiB	238	fits, \$807 headroom
Cloud Logging @ 1 KiB/request	2,630 GiB	1,290	\$5,244.58 — OVER by \$245
Cloud Logging @ 2 KiB/request	5,260 GiB	2,605	catastrophically over

At 2.758 billion requests/month, Cloud Logging at a realistic 1 KiB structured audit line costs **more than the entire remaining headroom**. Avoidable via Log Router exclusions plus a GCS sink — but it is a decision to be **made and stated**, not discovered on the first invoice.

Cost by posture

Non-hot-path lines (disks, registry, monitoring, audit = \$237) apply to every rung.

Posture	RPS	Hot-path	All-in	Fits \$5k?
A — scan every request (recommended)	1,064	3,737.58	3,974.58	yes — \$1,025 spare
A' — quota raised to 14 L4	1,862	4,742.76	4,979.76	yes, at the ceiling
B — risk-gated semantic ≥26%	4,151	5,000	5,237	no
C — deterministic only	7,200–9,400	5,000	5,237	no

The budget is not the binding constraint at the recommended point — NVIDIA_L4_GPUS = 8 is. File the quota increase before spending anything else: posture A' nearly doubles throughput for \$1,005/mo and is the best marginal purchase available.

Not included: customer BYOK model spend (out of envelope by definition) · the \$7 trtexec bake-off · Cloud SQL backup storage beyond the included allocation NF · support plan · any AWS spend during the migration overlap.

13 · What we must not claim

The list exists because several of these were about to be claimed.

LATENCY

1. **No "12 ms" without a posture, a prompt band, and a percentile.** Today's honest sentence: *"≈119 ms p50 with semantic classification on; a Tier-1-only posture is projected at 3–5 ms p50 once rewritten, with p99 unmeasured."*
2. **No streaming latency number from the current bench.** It measures the customer's model.
3. **No p99 for any posture.** The 40 ms clause is unbacked.
4. **No figure "on c4a / Axion."** Nothing was measured there.
5. **Not $N \approx 85$ RPS/host** — a 32–64-token number; at 1024 tokens it is ~5.
6. **Not 100k RPS.**

DETECTION

7. **We cannot claim the default posture detects prompt injection.** 3/10 same-family, **0/10 paraphrased.**
8. **We cannot claim a low false-positive rate.** 5/5 benign inline-code prompts and 3/12 ordinary tool descriptions hard-blocked at confidence 1.00, terminally.
9. **No tool/function-calling support at the 1024-token band** — ~6–7 ordinary tools tip the request into `block/dos/context_length_exceeded`.
10. **No supported prompt length above ~1,970 tokens** — ~800 in Hindi, ~750 in Tamil, in an India-only deployment.
11. **Streaming "block" is not a block.**
12. **Responses are not Tier-2 scanned end-to-end** — and telemetry attests that they are.
13. **No per-tenant policy isolation.**
14. **Policy enforcement is not fail-closed.**
15. **No bounded revocation or config-propagation time.** `auth:epoch` does not exist.
16. **No complete enforcement records.**
17. **Tier-2 does not see the whole response** — `_head_tail` silently drops the middle of any answer over 10,000 characters.
18. **No availability during cache maintenance** under the kill-switch contract as signed.

PROCESS

19. **Do not present target values as measurements.** The "2.39 ms" row is a sum of unbuilt targets, two of which the same document calls unreachable.
20. **potion-base-8M is not a viable Tier-2 replacement.** AUC 0.60 / 2.8% recall at the deployed window.
21. **The off-host option was not evaluated.** It was rejected on a price from a different cloud.

For the person who signs this

The 12 ms lock cannot be met with a semantic classifier on this hardware inside this budget, and no priced path to it exists. That part survived five adversarial reviews and a devil's advocate.

But the recommendation built on top of it is not safe to ship. It proposes defaulting to the deterministic layer and certifying 12 ms for that default — and **nobody measured what the deterministic layer detects until this review**: 0 of 10 paraphrased prompt injections caught, 5 of 5 benign questions containing inline code hard-blocked.

The plan spent seven agents optimising the latency of a control whose effectiveness had never been measured, and was about to publish an SLO for it. **Fix the instrument, measure the detection, price the GPU, then choose the posture — in that order.** The engineering work is real and ~10× is available this quarter with no new

dependencies. But the decision in front of you is a product decision about what this firewall actually stops, and it cannot be made from a latency table.

